

Prepared From Path Lesson

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Open on the anchor and get the learner curious about what will happen next. Activate the prior knowledge that butterflies come from eggs without yet naming the full cycle.

A nature center shows a time-lapse video of a monarch egg, caterpillar, chrysalis, and butterfly. The narrator says the video shows a life cycle. What does that mean?

- a. It shows stages an organism passes through, and adults can produce young again.
- b. It shows four animals competing for the same milkweed plant.
- c. It shows one animal learning to grow larger without changing form.
- d. It shows a path that ends when the adult butterfly appears.

A class raises painted lady butterflies. One student says the egg, caterpillar, chrysalis, and butterfly must be four different animals because they look nothing alike. Which response uses the idea of a repeating life cycle?

- a. They look different because they are stages of one organism's changing life.
- b. They are four animals that take turns using the same body parts.
- c. They are four animals, but the adult butterfly makes them look alike later.
- d. They are four animals that all come from eggs laid in the same place.

Build the core idea: the egg, caterpillar, chrysalis, and butterfly are stages of one organism's life cycle, and the cycle repeats when the adult butterfly produces young.

Stages of a monarch's life

STAGE OF LIFE CYCLE	WHAT THE MONARCH DOES
Egg	The tiny egg holds the young monarch inside.
Caterpillar	The caterpillar eats leaves and grows bigger.
Chrysalis	The chrysalis is a case that protects the monarch as it changes.

Butterfly	The adult butterfly flies and can lay eggs.
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A life cycle is the series of stages an organism passes through as it grows and develops. For a monarch butterfly, we can see this by following what happens from egg to adult. The cycle begins with an egg, then moves through the caterpillar and chrysalis stages, and finally becomes an adult butterfly.

Looking back at our monarch butterfly near the milkweed plant, we can see this pattern in action. The adult butterfly lands on a leaf and lays a tiny egg. That egg holds all the instructions needed for a new monarch to grow. The egg hatches into a caterpillar that spends its time eating milkweed leaves so it can grow bigger.

After the caterpillar grows enough, it forms a chrysalis around itself. Inside this protective case, the caterpillar's body changes into a completely different form. When the change is complete, an adult butterfly comes out. This adult can fly to find food and a mate.

The monarch's life cycle repeats when an adult butterfly produces young. When that adult butterfly lays eggs of its own, each egg starts the series of stages all over again. Each new generation passes through the same stages: egg, caterpillar, chrysalis, and butterfly.

Confront the two false models: that the egg, caterpillar, chrysalis, and butterfly are four different kinds of animals, and that the cycle stops at the butterfly.

A gardener finds small eggs on a leaf. A few weeks later, she sees caterpillars on the same plant, and later still, chrysalises. She asks why she never sees the eggs, caterpillars, and chrysalises standing next to each other. Which statement best answers her question?

- a. They are one organism changing from stage to stage, so one stage is replaced by the next.
- b. Each kind of animal avoids the others so it can find its own food.
- c. They do appear together, but only after the butterfly has stopped growing.
- d. Each stage hides underground until its turn in the life cycle arrives.

A student sees a caterpillar crawling on a stem. She says it must have been born a caterpillar and will have caterpillar babies. How would a correct understanding of a life cycle respond?

- a. The caterpillar is one stage in a repeating cycle of one organism's life.
- b. A caterpillar is a fully grown animal, so it cannot have babies of its own.
- c. Caterpillars are the only stage that can reproduce without becoming a butterfly.
- d. The caterpillar will remain a caterpillar because the cycle moves only forward.

Think about a path that has four steps: egg, caterpillar, chrysalis, butterfly. If the path simply stops after the butterfly, we are looking at a line that comes to an end. The butterfly is the last step, and then there is nothing new.

A life cycle is different because it does not stop at the butterfly. It still has the same four stages: egg, caterpillar, chrysalis, and butterfly. But after the butterfly, the path continues when the butterfly produces a new egg.

So, the line that stops goes egg, caterpillar, chrysalis, butterfly, then nothing. The life cycle goes egg, caterpillar, chrysalis, butterfly, and then back to a new egg.

Check whether the learner can use the idea of a repeating life cycle to say what must happen after the adult butterfly stage.

A class watches a single monarch butterfly lay an egg on a milkweed leaf. They know the egg will hatch and grow. What will happen after the butterfly that eventually emerges from this process is an adult?

- a. It can lay eggs of its own, starting the series of stages again.
- b. The cycle is complete, so the butterfly has no further role.
- c. It will search for the egg, caterpillar, and chrysalis that grew beside it.
- d. It will become a different kind of insect the following spring.